

ADE-Sentinel: Detecting and Extracting Adverse Drug Events from Medical Case Reports

Project Proposal · Department of Computer Science and Engineering
Khulna University of Engineering & Technology

Submitted by: Md. Sifatul Islam (2107xxx) · [Partner name] (2107xxx)

Submitted to: Dr. K. M. Azharul Hasan, Professor · Md. Nazirulhasan Shawon, Assistant Professor

What will be done

Adverse drug events - harmful reactions caused by a medication - are a major cause of hospitalisation, and most of the evidence about them sits in free text such as published case reports. A sentence mentioning a drug and a symptom does not thereby report one: “hepatotoxicity developed after methotrexate” and “methotrexate was given; no hepatotoxicity was observed” share almost every word. Separating them requires modelling negation, hedging, and word order.

We will build ADE-Sentinel, a two-stage pipeline. Stage 1 decides whether a sentence reports an adverse drug event and acts as a gate; Stage 2 runs only on the sentences it accepts and extracts which drug caused which effect as labelled spans. Three things will be delivered: embeddings trained on our own biomedical corpus, with evidence that they beat general-purpose vectors on this vocabulary; a benchmark of a model ladder over both stages, including how much Stage 2 degrades on predicted rather than gold input; and a simple demo. This is a methods study on published literature, not a clinical tool.

How it will be done

Data. ADE Corpus V2 (Gurulingappa et al., 2012), from MEDLINE case reports: ~23.5k labelled sentences for Stage 1 and ~6.8k with drug and effect character offsets for Stage 2. It ships with a single split and duplicate sentences, so we deduplicate and cut our own stratified 70/15/15 split at a fixed seed, shared by both stages. Separately, 100k–200k PubMed abstracts are fetched through the NCBI E-utilities API as an unlabelled domain corpus, with the query and date range recorded.

Preprocessing. A domain tokenizer preserving character offsets, since the Stage 2 labels are offsets that must become BIO tags; the converter is unit-tested, because off-by-one bugs fail silently. One sentence in five is positive, so imbalance is handled by metric choice.

Embeddings. The headline experiment: the word representation is the only variable, with tokenizer, splits, architecture, and seed held fixed. Four are compared - random, GloVe 300d, our Word2Vec, and our FastText (subwords should help on rare drug names) - under three kinds of evidence: vocabulary coverage by token and by type, nearest neighbours for a fixed probe list, and downstream F1 with only the embedding layer swapped.

Models. The ladder ascends in capacity, each rung a floor for the next. Stage 1: Naive Bayes on n-grams; logistic regression / SVM on TF-IDF; a BiLSTM with attention over each embedding matrix; BERT-base; and BiomedBERT, replaying the domain-versus-general comparison at the contextual level. Stage 2: BiLSTM softmax, BiLSTM-CRF, and BERT token classification, the CRF judged by the illegal tag sequences it removes.

Evaluation. Macro-F1 with per-class precision, recall, and a confusion matrix for Stage 1; entity-level seqeval F1 for Stage 2, not token accuracy, which the dominant O class inflates. Every run is logged with its commit, seed, and hyperparameters.

Analysis. A negation and hedging subset is selected programmatically, before any prediction is seen, and every Stage 1 model scored on it against the full test set. Error propagation is measured by running Stage 2 on gold positives and then on Stage 1's predicted positives, decomposing the entity-F1 drop.